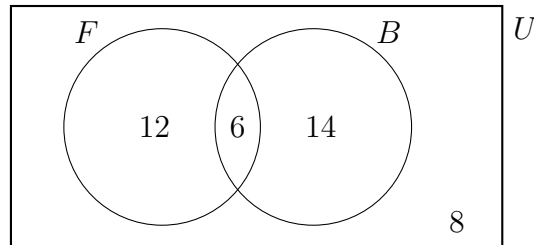


4.17 Quiz: Probability

1. In a group of 40 students, some play football (F) and some play basketball (B). Six students play both sports. This information is shown in the following Venn diagram.



- (a) Write down the number of students in the group who play football. [2]
- (b) One student from the group is chosen at random. Find the probability that
- (i) the student does not play football;
- (ii) the student plays football or basketball, but not both. [4]

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2. Events A and B are mutually exclusive with $P(A) = 0.55$ and $P(B) = 0.15$.

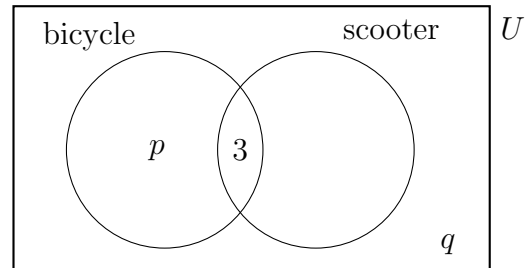
- (a) Write down $P(A \cap B)$. [1]
- (b) Find $P(A \cup B)$. [2]

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3. In a survey of 25 students, 18 own a bicycle and 7 own a scooter. Three students own both, as shown in the following Venn diagram. The values p and q represent numbers of students.



- (a) Find the value of p . [1]
- (b) Find the value of q . [1]
- (c) A student is chosen at random. Find the probability that the student owns a scooter but not a bicycle. [2]

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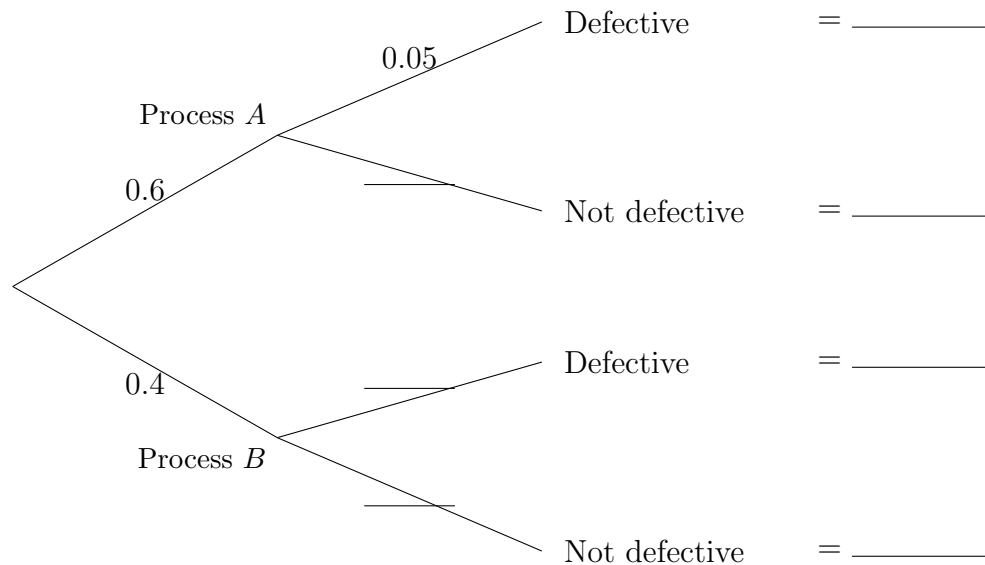
4. The following table shows the sports and snack preferences of 20 students.

	Peanuts	Popcorn	Total
Mets fans	4	10	14
Yankees fans	2	4	6
Total	6	14	20

- Find the probability that a student chosen at random is a Yankees fan who likes popcorn. [2]
- Find the probability that a student is a Mets fan, given that they like peanuts. [2]
- Determine whether being a Mets fan and liking peanuts are independent events. Justify your answer. [2]

5. A factory produces parts using two manufacturing processes. Process A produces 60% of all parts and process B produces the remaining 40%. The probability that a part from process A is defective is 0.05. The probability that a part from process B is defective is 0.15.

(a) Complete the tree diagram, showing all probabilities. [3]



- (b) A part is selected at random. Find the probability that the part is
- defective and from process A ; [2]
 - defective. [2]
- (c) Given that a part is defective, find the probability that it came from process A . [3]

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Name:

- drawer at random, one after the other, without replacement.

[2]

[4]